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clc; clear; close all;
bandwidth = 10e6;
rb_size = 180e3;
total_RB = floor(bandwidth / rb_size);
users = 1:6;
base = floor(total_RB / length(users));
alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users)))
+ 1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')

```

Total Resource Blocks:

```
disp(total_RB)
```

55

```
disp('RB Allocation per User:')
```

RB Allocation per User:

```
disp(alloc)
```

10      9      9      9      9      9

```

figure
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```

